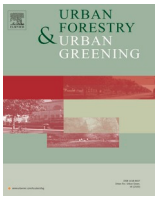
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Predicting citizens' participatory behavior in urban green space governance: Application of the extended theory of planned behavior

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ABSTRACT

The governance of urban green spaces (UGS) often involves a wide range of stakeholders, each with their own views, preferences, and needs. However, the traditional top-down governance of UGS has struggled to meet the needs of today's public. By reflecting the different needs of different social groups, public participation has become one of the innovative solutions used to solve the problems of public governance, which has been a widespread concern. Based on the theory of planned behavior (TPB), this study complements environmental responsibility (ER), environmental concern (EC), subjective environmental knowledge (SEK) and media influence (MI) as the factors influencing the formation of citizens' participatory attitude, and provides multiple influencing paths for increasing citizens' participatory intention and behavior. A structural equation model (SEM) was used to test the hypotheses of the newly constructed theoretical model based on 1191 questionnaires administered face-to-face in Beijing. The results demonstrate that perceived behavioral control has the greatest positive influence on citizens' intentions and behaviors; that is, their intention and behavior to participate is strongly restricted by the objective conditions of the citizens themselves. Among the additional impact factors, expected personal gratification (EPG) also has a positive effect on intention. Local governments can increase citizens' intentions and behaviors to participate in UGS governance through two measures: one is to optimize the process of citizens' participation in UGS governance to reduce the threshold of participation; the other is to enhance the city's infrastructure and cultural and entertainment activities to enhance citizens' environmental responsibility.

1. Introduction

According to a United Nations report released in 2018, 55 % of the world's population currently live in cities, and this number is expected to increase to 68 % by 2050, putting higher demands on urban planning and public services (United Nations Department of Economic and Social Affairs, 2018). Urban green spaces (UGS) include public space entities such as parks, grasslands, and forests in urban landscapes, and provide a variety of environmental and social services, such as air and water purification and entertainment opportunities (Bowler et al., 2010). Based on data from the National Bureau of Statistics of China (2018), China has an urban population of approximately 831.4 million (approximately 59.6 % of the total population) and an urban green space area of 3.05 million hectares (including 723,740 ha of urban park green space). These figures show that the per capita urban green space with open space function in China is approximately 8.70 square meters. Moreover, Beijing, as the capital of China, has a green coverage rate of 48.44 %, ranking first of all provinces in China, with a per capita urban park green area of 16.3 square meters (Office of Beijing Greening Commission, 2018). The highly concentrated population brings higher governance requirements to the urban environment.

As public places in cities, UGS serve a variety of citizens whose perceptions, preferences, and demands also vary (Janse and Konijnendijk, 2007). As a result, governance of UGS, whether through planning, design, or management, is usually very complex, and involves a wide range of stakeholders (Van Herzele et al., 2005). The traditional top-down or bureaucratic mode of public governance has been increasingly criticized as inappropriate (Jim and Chen, 2006a; Huang, 2010), because it is mainly based on professional assumptions and not on the needs of the public (Jim and Chen, 2006a). In the process of

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traditional governance, the public is regarded as a passive receiver, which makes the results of governance arbitrary, even outdated, failing to reflect the different needs of different social groups, including their desire to participate in relevant decisions (Sanesi and Chiarello, 2006).

Public participation has been emphasized as an innovative solution to address public governance issues (Ward Thompson, 2002). It has received a great deal of attention in urban forest governance in recent years (Gulsrud et al., 2018). Although it is difficult to accurately measure the effect of public participation, the general conviction is that such participation will lead to better policy decisions and more public support for government policies (Karlsson et al., 2012). Public participation could provide a valuable opportunity for people to express their opinions and needs, so that relevant decisions could better meet their different needs. Compared with passive acceptance of information-based environmental education, this kind of active environmental participation can be regarded as an experiential form of lifelong learning, which is very effective in changing people's behavior toward the natural environment (Wallington et al., 2005). More importantly, through this participation, the public can gain a sense of control and empowerment, and further enhance their citizenship and willingness to participate again (Shan, 2012). For example, in Japan, citizens participated in the design and evaluation of urban parks, which enabled them to better use or even take care of urban parks (Kobori and Primack, 2003). In Finland, both the green space planning authorities and participating citizens found the participatory green space design method to be useful (Sipilä and Tyrväinen, 2005). An empirical study in Turkey also showed that effective cooperation between the competent authorities of urban forestry and other public institutions, universities, local civil groups, and non-governmental organizations could enhance the effect of governance in urban forestry (Atmis, 2016).

In China, public participation has increasingly been required by the government in dealing with public issues (Yang, 2008; Zhong and Mol, 2008). The Law of the People's Republic of China on Environmental Protection, which came into force in 2015, has set out a new chapter for public participation, which has clarified the legal status of public participation in environmental protection in China. However, many citizens do not always desire to exercise their civil rights (e.g., public participation) despite having protection from the law (Lee et al., 2019). In the past few decades, practical research has shown that public participation in China has been inefficient (Shan, 2012). Apart from other reasons such as the imperfection of the original legal system (Yang, 2008), the public's willingness to participate and their attitude to so doing is a crucial factor (Shan, 2012). As the social basis of public participation, the willingness and attitude of public participation could affect the success of participatory decision-making in related fields (Wall et al., 2006). However, previous research on citizens' willingness and attitude to participate in UGS governance has mainly focused on qualitative analysis. Most relevant studies focus on the political and institutional aspects (Jho and Song, 2015) or micro case studies of citizen participation, such as community participation in green space governance research (Glover et al., 2005; Rosol, 2010; Hoffman, 2019). Moreover, quantitative studies are insufficient, particularly research on the influencing factors and interaction mechanisms of participatory behavior. Understanding these factors and how they affect participation is essential to promoting and sustaining participation (Fors et al., 2019).

To fill these gaps in the literature, we conducted a face-to-face questionnaire survey in Beijing to collect data on citizens' participatory behavior in UGS governance. The possible contributions of this study are mainly manifested in two aspects. First, from the perspective of citizens, considering the individual influencing factors of participatory behavior, and the interaction mechanism between these factors, can supplement the lack of empirical research. Second, the theory of planned behavior is expanded. Based on previous studies, the most susceptible attitude latent variables were selected to refine the influence path, so that the results of the model could be more operable for policymakers. The remainder of this paper is organized as follows. Section two comprises a literature review and conceptual framework, which mainly summarizes previous studies on citizens' participation in UGS governance behavior, as well as the specific construction of the conceptual model of this study. Section three is the methods, which mainly describes the collection of sample data and the specific method of conceptual model implementation. Section four is the results, which show the reliability and validity of the model and the test results of the hypothesis. Section five is the discussion, which mainly discusses the core findings and limitations of the research. Section six presents the conclusions.

2. Literature review and conceptual framework

2.1. Factors influencing citizen participation in environmental governance

There is a wide range of literature on the factors that promote citizen participation in environmental governance. For example, Mattijssen et al. (2017) conducted document analysis and interviews with three European cases characterized by long-term involvement of citizens in place-keeping. They identified three factors of particular relevance. The first is the degree of formalization, which indicates the established rules and procedures within groups. The second is the adaptive capacity of citizens, which means they need to be able to adapt to external changes in politics, economics, and culture. The last and most important factor is the support of the local governments. Local governments can support participation by providing security via stable policies, allowing long-term management contracts, etc. The first and final factors mentioned here are basically external factors. Previous studies have also shown that residents' participation behavior could be affected by support and partnership from authorities (Mathers et al., 2015; Jorgensen et al., 2007). Particularly for smaller community groups, without the continuous support of local authorities, they may disappear due to the lack of internal capacity and external support (Mathers et al., 2015). Therefore, the impact on participation of external factors should be included in the theoretical framework. The second factor regards citizens' cultural literacy, which indicates that some internal characteristics of citizens are likely to impact their participatory behavior.

In fact, previous studies have found that age seems to impact participatory behavior, although the results are unclear. Still and Gerhold (1997) conducted an email survey of volunteer organizations in New York and Philadelphia and found that nearly half of urban forestry volunteers (43 %) were between 35 and 50 years old. Similarly, a study by Straka et al. (2005) in South Carolina (United States) showed that young people are more likely to participate in urban and community forestry projects and that most participants are between 30 and 49 years old. By contrast, according to Wall (2006), working people, that is, people aged between 18 and 65, are less involved in such projects than children, adolescents, and the elderly. A survey conducted by the National Voluntary and Charitable Survey in the UK showed similar results, with 42 % of volunteers in the conservation, environmental, heritage, and animal welfare sectors over 55 years old (Low et al., 2007). To some extent, age can more or less reflect intrapersonal constraints, such as their financial resources, time, physical condition, etc. However, according to the results of previous studies, the direct use of age as the main factor of participation behavior may not be accurate, and may vary based on national policy. Therefore, intrapersonal constraint is relatively more suitable than age for describing such influencing factors (Gage and Thapa, 2012).

In addition to these social factors, the possible motivations for voluntary participation in English-speaking countries include the desire to improve and help the local environment (Mathers et al., 2015), fascination with the environment or with a place (Miles et al., 1998), which is meaningful for society, future generations, or communities (Measham and Barnett, 2008). Sloane and Pröbstl-Haider (2019) found that the most important motivational factors were "helping the environment" and "learning natural knowledge" by comparing the

motivations of voluntary participation in the UK and Austria. Compared with other factors, the sample differences of these two factors between countries are minor. This means that, despite cultural differences, people's desire to improve the environment and to have access to the natural environment remains unchanged, and this has become a major driver of their participation in environmental governance. These factors can be classified to the participants as attitudes, knowledge, preferences, etc., which will be further discussed in the theoretical model.

2.2. Theory of planned behavior

Research within the environmental psychology and sociology fields has focused on rational-choice models, which have been used to explain people's decision-making process in terms of environmental behavior (Gao et al., 2017). The theory of planned behavior (TPB) is derived from the theory of reasoned action (TRA) (Ajzen, 1985), which is widely used to predict or explain people's actual behavior through behavioral intention (Martin et al., 2017). According to TPB, presented in Fig. 1, attitude toward behavior, subjective norm, and perceived behavioral control are the three influencing factors that determine behavior intention, while behavioral intention and perceived behavioral control are the direct factors that determine behavior (Ajzen, 1991). Attitude can be defined as an individual's evaluation result after weighing the advantages and disadvantages of the specific behavior (De Jong et al., 2019). Subjective norm indicates the influence of the social environment around an individual on their specific behavior (Wang et al., 2019). For example, relatives, friends, neighbors, or even the government could have an impact on an individual's behavior. Perceived behavioral control refers to the difficulty that an individual perceives in terms of implementing a specific behavior (Ajzen, 1991). These include their own financial resources, time constraints, or lack of professional knowledge.

TPB has been widely used in various fields, whenever it is assumed that people can consciously decide their behavior (De Jong et al., 2019). Specifically, participatory behavior in UGS governance is a kind of environmental participatory behavior. Although there are few studies that use TPB to quantitatively study citizen participation in UGS governance, combined with the application of TPB in fields related to public participation behavior and pro-environment behavior (De Jong et al., 2019; Gkargkavouzi et al., 2019), it can also provide a good reference for the theoretical development of this study. According to the previous analysis, citizens' participation in UGS governance may be affected by external factors, personal ability, and attitude, which is similar to the subjective norm, perceived behavioral control, and attitude in TPB. Moreover, the behavior of participating in UGS governance belongs to the behavior that needs to be controlled by consciousness, not impulsive or habitual behavior, which conforms to the premise of TPB use (De Jong et al., 2019). Therefore, it is feasible to choose TPB as the basic theoretical model in this study, and the subsequent extension will be carried out on this basis. According to TPB theory and the literature review above, the following hypotheses are proposed:

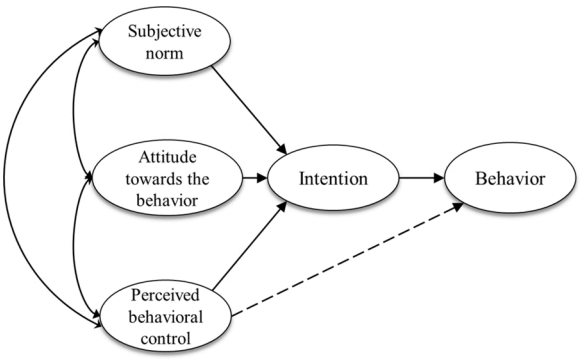


Fig. 1. Theory of planned behavior (Ajzen, 1991).

H1. Citizens' attitude toward participation in UGS governance has a positive impact on their intention to participate.

H2. Perceived behavioral control has a positive impact on citizens' intention to participate in UGS governance.

H3. Subjective norm has a positive impact on citizens' intention to participate in UGS governance.

H4. Citizens' intention has a positive impact on their participatory behavior in UGS governance.

H5. Perceived behavioral control has a positive impact on citizens' participatory behavior in UGS governance.

2.3. Research model

To study the determinants of citizen participation in UGS governance, we combine TPB with previous research on citizen participation and pro-environmental behavior. This means that we need to distinguish some similar concepts in the previous research, which is the basis of the expansion of our research model.

2.3.1. Environmental responsibility (ER)

Environmental responsibility is defined as a psychological state that reflects the degree to which citizens feel compelled to take beneficial actions toward the environment (Punzo et al., 2019). In fact, people with a sense of responsibility may think that they should attempt to improve the existing deficiencies rather than transferring this responsibility to others, which makes them more inclined to take relevant practical actions (Bateman and O'Connor, 2016). In other words, the more citizens feel personally responsible toward the environment, the more likely they are to take relevant actions to protect the environment (Punzo et al., 2019). A study on responsibility, concern, and attitude conducted by Verma et al. (2019) in the field of environmental protection showed that the sense of responsibility can enhance the intention to visit green hotels by improving people's environmental concerns and attitudes. Similarly, in this study, we attempt to verify how citizens' environmental responsibility affects their intention to participate and their behavior when they are not satisfied with the state of the current UGS. This is of great significance to the government that wants citizens to participate more in environmental governance, because the verification of this part of the hypothesis can provide a reference path to enhance citizens' intention to participate. Based on the above analysis, the following hypothesis is proposed:

H6. Citizens' sense of environmental responsibility has a positive impact on their attitude toward participation in UGS governance.

H7. Citizens' sense of environmental responsibility has a positive impact on their environmental concerns.

2.3.2. Environmental concern (EC)

The widely accepted definition of environmental concern provided by Dunlap and Jones (2002) refers to the extent to which people are aware of environmental problems and support efforts to solve them. Most studies have confirmed through empirical research that environmental concern increases the intention to engage in pro-environmental behavior (Verma et al., 2019). On the contrary, other studies have shown that environmental concern does not directly affect behavior, particularly in developing countries (Harris, 2006). Bamberg (2003) considered that the reason for the controversy seems to be the lack of description of the intermediate variables between environmental concern and behavior. When people are concerned about certain environmental problems, one possibility is that they will actively seek relevant environmental knowledge to fill their knowledge gaps. This part of the hypothesis is part of the path whereby environmental responsibility

affects intention, and also expands a possible path to improving citizens' intentions to participate. Hence, it can be assumed that

H8. Environmental concern has a positive impact on subjective environmental knowledge.

2.3.3. Subjective environmental knowledge (SEK)

As a very important variable, many scholars consider environmental knowledge a prerequisite for fostering environmental attitudes (Oreg and Katz-Gerro, 2006), and environmental attitude is the core intermediate variable between environmental knowledge and behavior (Polonsky et al., 2012). Environmental knowledge can be divided into two categories: subjective and objective knowledge (Pagiaslis and Krontalis, 2014; Vicente-Molina et al., 2013). Subjective knowledge refers to specific knowledge related to actions, which shows how much a person knows about specific environmental actions (Pagiaslis and Krontalis, 2014). Objective knowledge is defined as accurate information about the environment stored in a person's long-term memory (Vicente-Molina et al., 2013). In this study, we use the definition of subjective environmental knowledge, because previous studies have shown that subjective environmental knowledge may be more predictive of environmental behavior than objective knowledge (Ellen, 1994). This hypothesis is an important link between environmental responsibility, environmental concern, and environmental attitude. Based on the above analysis, it can be assumed that:

H9. Subjective environmental knowledge has a positive impact on citizens' attitudes toward participation in UGS governance.

2.3.4. Media influence(MI)

Information is an essential element of environmental governance, without which there can be no qualified public communication process or effective public participation (Mazzarino et al., 2020). The media usually play a crucial role in delivering important information to people, particularly when it comes to environment-related issues (Yu et al., 2017). Many previous studies have found that social media participation significantly affects people's cognitive stages, including awareness and knowledge (Hutter et al., 2013). Testing this part of the hypothesis can provide a more feasible operation path for improving citizens' intentions to participate. Thus, it can be assumed that:

H10. The media has a positive impact on citizens' access to knowledge of participation in UGS governance.

2.3.5. Expected personal gratification (EPG)

Expected personal gratification refers to the emotional satisfaction that citizens can expect to attain from participating in UGS governance before participating. Fun or enjoyment is one of the most attractive factors when explaining why someone spends a lot of time on participatory projects (Luthiger and Jungwirth, 2007). Research on open government projects has also proved that fun is one of the main motivations for citizen participation (Wijnhoven et al., 2015). Other studies on citizen participation have also demonstrated that expected personal gratification can be one of the driving forces behind intention to participate (De Jong et al., 2019). In this part of the hypothesis, we attempt to test the impact of expected personal gratification on citizens' intentions to participate in UGS governance. Therefore, it can be proposed that:

H11. Citizens' expected personal gratification has a positive impact on their intention to participate in UGS governance.

Based on the above discussion, we propose a comprehensive model based on the TPB model enriched with the addition of environmental responsibility, environmental concern, subjective environmental knowledge, media influence, and expected personal gratification variables. Fig. 2 shows the final hypothetical relationship of this study. This theoretical model aims to explain the influence of some psychological and external factors on citizens' intention and behavior when participating in UGS governance.

3. Methods

3.1. Questionnaire design

Our research team has been monitoring the public's attitude towards UGS in Beijing for five years. The questionnaire is the result of adjustment based on the feedback of several surveys. The questionnaire has three sections. The first part is an introduction to the questionnaire, which explains the purpose of the survey and how to answer each question. The second part includes participants' demographic information, such as their gender, age, education level, and occupation. The final part is the core part of the questionnaire, which proposes questions based on the theoretical model used in this study. In this part, all the latent variables were measured with multiple indicators, mostly adapted from previous literature. Some indicators were modified to fit the current research setting. Responses were provided on a five-point Likert scale (1=strongly disagree; 5=strongly agree). Following Hinkin (1998) guidelines, to enable respondents to understand the subject more effectively, all items were written simply and concisely. In addition, while some previous studies used a mixed measurement of positive and negative items, in our study, to facilitate respondents' understanding of the items, all items are described in a positive way, rather than as a mixture of positive and negative items. The relevant items and their references are given in Table 1.

3.2. Sample & data collection

To collect data to test the proposed hypotheses, a questionnaire survey was conducted in the urban area of Beijing in November 2019. A total of 1400 questionnaires were distributed to Beijing residents by

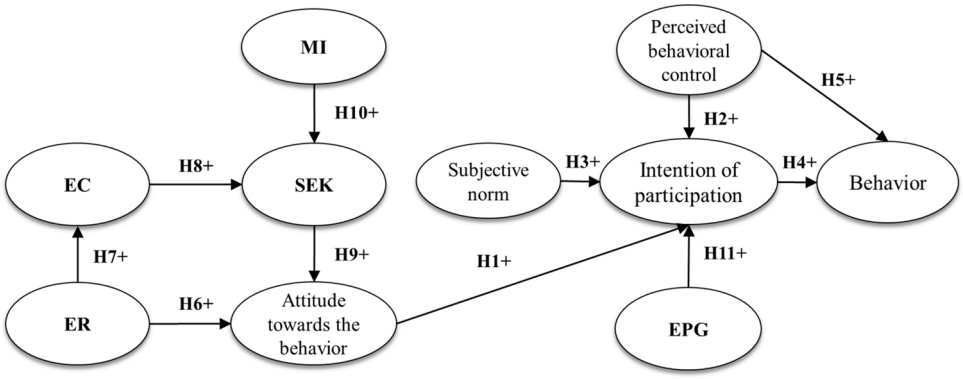


Fig. 2. Research model.

Table 1
Items and references.

Latent variable	Observed variable	Items	References
Attitude toward the Behavior (AB)	AB1	I think it's good for citizens to participate in urban green space governance.	Ajzen, 1991; Wang et al., 2018
	AB2	I believe that citizen participation in urban green space governance is worthwhile.	
	AB3	I believe that citizen participation in urban green space governance is wise.	
Subjective Norm (SN)	SN1	My family members think I should participate in urban green space governance.	Ajzen, 1991; Taufique and Vaithianathan, 2018
	SN2	My friends think I should participate in urban green space governance.	
	SN3	My neighbors think I should participate in urban green space governance.	
Perceived Behavioral Control (PBC)	PBC1	I have the right skills to participate in urban green space governance.	De Jong et al., 2019; Meng and Cui, 2020
	PBC2	I have sufficient time to participate in urban green space governance.	
	PBC3	I am well-equipped to participate in urban green space governance.	
Intention of Participation (I)	I1	I would certainly think about participating in urban green space governance.	Ajzen, 1991; De Jong et al., 2019
	I2	I think I would like to participate in urban green space governance.	
	I3	I will participate in urban green space governance if I have an opportunity.	
Behavior (B)	B1	I have participated in volunteer activities in urban green space recently.	Ajzen, 1991; Sloane and Pröbstl-Haider, 2019
	B2	I have reported violations of urban green space recently.	
	B3	I have participated in the policy consultation of urban green space recently.	
Expected Personal Gratification (EPG)	EPG1	Participating in urban green space governance seems fun to me.	Wijnhoven et al., 2015; De Jong et al., 2019
	EPG2	Participating in urban green space governance would give me a feeling of satisfaction.	
	EPG3	Participating in urban green space governance would give me a sense of empowerment.	
Environmental Responsibility (ER)	ER1	I have an obligation to maintain urban green space.	Onwezen et al., 2013; Verma et al., 2019
	ER2	I have an obligation to save resources and protect the environment.	
	ER3	I feel guilty if I waste resources or pollute the environment.	
Environmental Concern (EC)	EC1	I am concerned about the occupation and destruction of urban green space.	Lee, 2008; Fu et al., 2020
	EC2	I am concerned about resource waste and environmental pollution.	
	EC3	I care about the issues related to the urban ecosystem.	
Subjective Environmental Knowledge (SEK)	SEK1	I know how to participate in urban green space governance.	Fu et al., 2020
	SEK2	I know about several ways to participate in urban green space governance.	
	SEK3	I can participate in urban green space governance through the network.	
	SEK4	Citizen participation contributes to the harmonious development of urban green space.	
Media Influence (MI)	MI1	The media is a good source of knowledge regarding how to participate in urban green space governance.	Rahbar and Wahid, 2011; Trivedi et al., 2018
	MI2	The media has shown me the benefits of participating in urban green space governance.	
	MI3	The media has enhanced my interest in participating in urban green space governance.	

stratified and convenience sampling. The questionnaire was designed to be anonymous. Each investigator explained the purpose of the questionnaire to the participants and obtained their consent. First, in terms of spatial location, the central urban areas (including Dongcheng District, Xicheng District, Chaoyang District, Fengtai District, Shijingshan District, and Haidian District) were divided into six groups. Then, large open spaces such as parks, residential green spaces, and commercial squares were randomly selected in each urban area for the questionnaire survey, and the number of questionnaires in each urban area was controlled so that the final collected questionnaire samples were approximately evenly distributed in each group. To ensure the quality of the data, all research team members were trained to appropriately answer any questions raised by the respondents. After eliminating invalid questionnaires with incomplete filling and consistent scoring of all options, 1191 questionnaires remained, with an effective response

rate of 85.07 %. We follow Kline's (2011) recommendation that the sample size of SEM should be 10 times that of the observed indicators. As there are 31 items in this study, at least 310 samples are required. Therefore, our final sample size of 1191 was sufficient for this study.

The collected demographic data and their coding in the questionnaire are shown in Table 2. They mainly have the following characteristics: (1) The sample sizes of males and females were roughly the same. (2) More than half of respondents were predominantly young, aged between 18 and 39. (3) More than half of respondents had a bachelor's degree or above. (4) There was a relatively high number of married persons among the respondents. (5) Most of the respondents were company employees. (6) Most respondents had a monthly disposable income of below 12,000 RMB. In general, the respondents in this survey covered a wide range of social groups, and most social groups had a relatively adequate sample size, which was highly representative and conducive to the follow-up study.

3.3. Analysis method

Structural equation modeling (SEM) is a powerful multivariate statistical modeling technology that is used to test the theoretical model and hypothesis (Kline, 2011). Compared with other statistical techniques such as multiple regression, SEM can measure the latent constructs that cannot be observed directly, and can simultaneously analyze the relationship and intensity between multiple observed variables and latent variables. Therefore, this study used a structural equation model to test the proposed comprehensive model and the hypothesis relationship between variables.

SEM consists of two parts: the structural model, which is used to describe the causal relationship between latent independent variables and latent dependent variables; and the measurement model, which is used to describe the linear relationship between latent variables and observed variables. There are two common forms of measurement equation, which can be expressed as follows:

$$x = \Lambda_x \xi + \delta \quad (1)$$

$$y = \Lambda_y \eta + \varepsilon \quad (2)$$

where x are the column vectors of the exogenous variables; Λ_x are the factor loading matrices that associate the latent exogenous variables and observed variables; y are the column vectors of the endogenous variables; and Λ_y are the factor loading matrices that correlate the latent endogenous variables and observed variables. ξ and η represent the latent exogenous and latent endogenous variables, respectively. δ and ε are error terms.

The structural equation can be written as follows:

$$\eta = B\eta + \Gamma\xi + \zeta \quad (3)$$

where η and ξ are the column vectors of the endogenous and exogenous latent variables, respectively. B and Γ are factor loading matrices. ζ are error terms.

SPSS 25.0 and AMOS 22.0 were used to analyze the reliability and validity of the questionnaire data. More details can be found in the results section.The normality test of samples showed that the CR value of the multivariate kurtosis coefficient was 177.18 > 1.96 (i.e., P > 0.05), indicating that the multivariate data of the sample conforms to a normal distribution. Therefore, it is feasible to use the maximum likelihood estimation (MLE) in model estimation.

Table 2
Demographic characteristics of the respondents.

Characteristics	Category	Frequency	Percentage (%)
Gender	1 = Male	579	48.61
	2 = Female	612	51.39
Age	1 = 17 or under	36	3.02
	2 = 18–29	423	35.52
	3 = 30–39	280	23.51
	4 = 40–49	163	13.69
	5 = 50–59	134	11.25
	6 = over 60	155	13.01
Education	1 = Primary school or below	29	2.43
	2 = Junior high/Technical secondary school	133	11.17
	3 = Senior high school/ Junior college	284	23.85
	4 = Undergraduate	498	41.81
	5 = Master or above	247	20.74
Marital Status	1 = Unmarried	448	37.62
	2 = Married	688	57.77
	3 = Other	55	4.61
Occupation	1 = Farmer	11	0.92
	2 = Worker	33	2.77
	3 = Company employee	452	37.95
	4 = Staff of public institutions	170	14.27
	5 = Civil servant	62	5.21
	6 = Self-employed	48	4.03
	7 = Freelance	50	4.20
	8 = Student	137	11.50
	9 = Retirement	170	14.27
	10 = Active serviceman	4	0.34
	11 = Other	28	2.35
	12 = Unemployed	26	2.18
Monthly disposable income	1 = 4000 RMB and below	306	25.69
	2 = 4001–8000 RMB	390	32.75
	3 = 8001–12000 RMB	271	22.75
	4 = 12,000–16000 RMB	94	7.89
	5 = 16,000–20,000 RMB	42	3.53
	6 = 20,000–24,000 RMB	18	1.51
	7 = 24,000–28,000 RMB	30	2.52
	8 = 28,000–32,000 RMB	19	1.60
	9 = 32,000–36,000 RMB	6	0.50
	10 = 36,000 RMB and above	15	1.26

4. Results

4.1. Reliability and validity testing

A confirmatory factor analysis (CFA) was conducted to assess the reliability and validity of the construct measures. Cronbach's alpha and composite reliability (CR) were used to measure the reliability of each construct in the questionnaire. As shown in Table 3, the Cronbach's alphas of all constructs ranged between 0.851 and 0.944, exceeding the standard of 0.70 (Fu et al., 2020). This indicates that all constructs in the questionnaire are internally consistent and reliable. As Cronbach's alpha tends to underestimate reliability in small sample sizes, some studies suggest using CR as a supplement (Hair et al., 2016). All CR values presented in Table 3 are between 0.853 and 0.943, all above the standard of 0.70 (Fornell and Larcker, 1981), indicating a high degree of internal consistency among the latent variables.

Convergent and discriminant validity were used to evaluate the construct validity of the model. In general, convergent validity is assessed by factor loading and average variance extraction (AVE). Table 3 shows that the standardized factor loadings of all items exceed the suggested threshold of 0.50 (Chin, 1998) and are statistically significant ($p < 0.001$). The AVE values are between 0.606 and 0.847, higher than the recommended threshold value of 0.50 (Fornell and Larcker, 1981). This indicates that each observed variable has high explanatory power for their latent variable. The values of the square root of AVE are greater than the inter-construct correlations (Table 4), while low inter-construct correlations ($r < 0.85$) (Hair et al., 2017) are indicative of great discriminant validity between different constructs. In conclusion, these values indicate that the measurement model has high reliability and validity.

Table 3
Reliability and validity test results.

Latent variable	Observed variable	Mean	SD	Standardized factor loading	CR	AVE	Cronbach's alpha
Attitude toward the Behavior (AB)	AB1	4.00	0.95	0.841***	0.912	0.775	0.912
	AB2	3.94	0.99	0.901***			
	AB3	3.93	1.00	0.898***			
Subjective Norm (SN)	SN1	2.83	0.97	0.739***	0.899	0.749	0.894
	SN2	2.74	0.99	0.952***			
	SN3	2.64	1.04	0.892***			
Perceived Behavioral Control (PBC)	PBC1	3.30	1.05	0.877***	0.880	0.710	0.881
	PBC2	3.50	1.03	0.839***			
	PBC3	3.46	1.03	0.810***			
Intention of Participation (I)	I1	3.74	0.93	0.872***	0.909	0.768	0.911
	I2	3.71	0.92	0.905***			
	I3	3.61	0.98	0.852***			
Behavior (B)	B1	2.94	1.23	0.940***	0.943	0.847	0.944
	B2	2.97	1.22	0.910***			
	B3	2.89	1.25	0.910***			
Expected Personal Gratification (EPG)	EPG1	4.11	0.90	0.832***	0.853	0.660	0.851
	EPG2	3.96	0.94	0.745***			
	EPG3	4.10	0.88	0.856***			
Environmental Responsibility (ER)	ER1	4.13	0.88	0.865***	0.865	0.683	0.859
	ER2	4.14	0.88	0.904***			
	ER3	4.15	0.84	0.695***			
Environmental Concern (EC)	EC1	3.37	1.19	0.828***	0.872	0.696	0.869
	EC2	3.65	1.24	0.911***			
	EC3	3.57	1.22	0.756***			
Subjective Environmental Knowledge (SEK)	SEK1	3.78	0.98	0.920***	0.857	0.606	0.867
	SEK2	3.75	0.97	0.674***			
	SEK3	3.70	1.02	0.859***			
	SEK4	4.06	0.91	0.620***			
Media Influence (MI)	MI1	3.79	0.92	0.896***	0.919	0.790	0.921
	MI2	3.84	0.89	0.893***			
	MI3	3.77	0.92	0.878***			

Note: Significance level *** $p \leq 0.001$.

Table 4
Correlations between the constructs.

Correlation	AB	SN	PBC	I	B	EPG	ER	EC	SEK	MI
AB	**0.880**									
SN	0.302***	**0.866**								
PBC	0.238***	0.470***	**0.842**							
I	0.325***	0.326***	0.444***	**0.877**						
B	0.029	0.228***	0.497***	0.380***	**0.920**					
EPG	0.133***	0.141***	0.119***	0.151***	0.056	**0.812**				
ER	0.214***	0.128***	0.089**	0.308***	0.002	0.330***	**0.826**			
EC	0.229***	0.137***	0.129***	0.171***	0.053	−0.018	0.117***	**0.834**		
SEK	0.265***	0.322***	0.227***	0.236***	0.146***	0.085**	0.128***	0.175***	**0.778**	
MI	0.386***	0.368***	0.412***	0.540***	0.258***	0.145***	0.297***	0.186***	0.266***	**0.889**

Note: The diagonal (in bold) shows the square root of the AVE; significance level * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$.

4.2. Structural model and hypothesis testing

Indices such as χ^2/df, GFI, PGFI, TLI, CFI, IFI, SRMR, and RMSEA were selected to test the goodness of fit. The results are as follows: $1 < \chi^2/df = 2.305 < 3$ (suggested threshold), GFI = 0.955 > 0.90, PGFI = 0.726 > 0.50, TLI = 0.977 > 0.90, CFI = 0.981 > 0.95, IFI = 0.981 > 0.90, SRMR = 0.046 < 0.05, RMSEA = 0.033 < 0.08 (Liu et al., 2017). These indices are all within the recommended values, indicating that the

Table 5
Hypotheses' test results.

Hypothesis	Standardized path coefficient (β)	t-value	Result	Direct effect	Indirect effect	Total effect
ER → EC(+)	0.125	3.880***	Supported	0.125	—	0.125
EC → SEK(+)	0.102	3.331***	Supported	0.102	—	0.102
MI → SEK(+)	0.206	6.625***	Supported	0.206	—	0.206
ER → AB(+)	0.198	6.464***	Supported	0.198	0.002	0.200
SEK → AB(+)	0.147	4.949***	Supported	0.147	—	0.147
SN → I(+)	0.060	1.875	Rejected	0.060	—	0.060
PBC → I(+)	0.433	12.088***	Supported	0.433	—	0.433
EPG → I(+)	0.082	2.855**	Supported	0.082	—	0.082
AB → I(+)	0.145	4.898***	Supported	0.145	—	0.145
I → B(+)	0.202	6.168***	Supported	0.202	—	0.202
PBC → B(+)	0.394	12.010***	Supported	0.394	0.087	0.481

Note: significance level * $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$.

model fits the data well.

Table 5 shows the results of the hypothesis test and the direct and indirect effect measurements. After removing the insignificant path (hypothesis 3), the final revised model is shown in Fig. 3. The supporting condition of the hypothesis is that the absolute t-value exceeds 1.96, that is, $p < 0.05$ (Hair et al., 2016). Apart from one hypothesis, all the others satisfy the supported conditions. In the basic model of TPB, perceived behavioral control ($\beta = 0.394$, $t = 12.010$; $p < 0.001$) was found to have the greatest significant positive effect on the participating behavior, even greater than the participating intention ($\beta = 0.202$, $t = 6.168$; $p < 0.001$). Similarly, compared with attitude toward behavior ($\beta = 0.145$, $t = 4.898$; $p < 0.001$), perceived behavioral control ($\beta = 0.433$, $t = 12.088$; $p < 0.001$) had a stronger positive significant effect on participating intention. However, subjective norm ($\beta = 0.060$, $t = 1.875$; $p > 0.05$) failed to influence participation intention. Among the newly added factors, expected personal gratification ($\beta = 0.082$, $t = 2.855$; $p < 0.01$) had a significant positive effect on participating intention, albeit relatively weak. Environmental responsibility ($\beta = 0.198$, $t = 6.464$; $p < 0.001$) and subjective environmental knowledge ($\beta = 0.147$, $t = 4.949$; $p < 0.001$) have significant positive effects on attitude toward behavior, respectively. Environmental responsibility ($\beta = 0.125$, $t = 3.880$; $p < 0.001$) also has a significant positive effect on environmental concern. Environmental concern ($\beta = 0.102$, $t = 3.331$; $p < 0.001$) and media influence ($\beta = 0.206$, $t = 6.625$; $p < 0.001$) both had significant positive effects on subjective environmental knowledge, among which media influence has nearly twice as much effect on subjective environmental knowledge as environmental concern.

Table 4 depicts the direct and indirect effects among the latent variables in the structural model. Bootstrap was used to test the significance of direct and indirect effects, with a confidence interval of 95 % and 5000 samples. The significant criterion is that the interval between the upper and lower limits of the estimate does not contain zero value. According to the test results, environmental responsibility has a significant direct effect (0.138–0.261) and indirect effect (0.001–0.004) on attitude toward behavior, which indicates the existence of a partial mediating effect in the ER-EC-SEK-Attitude path. Similarly, perceived behavioral control has a significant direct effect (0.318–0.466) and indirect effect (0.058–0.125) on behavior, which indicates the existence of a partial mediating effect in the PBC-I-Behavior path.

5. Discussion

5.1. Influencing factors of intention and behavior to participate

The purpose of this study was to explore how citizens' individual factors affect their UGS governance behavior and what demographic characteristics affect these factors. In the first part, we proposed 11 hypotheses, including two mediating effects, which were tested by SEM. Except for H3 (SN→I), which failed to pass the test, the other 10 hypotheses were statistically significant. In fact, Ajzen (1991) noted that in TPB, the relationship between subjective norms and intentions was weak. In previous studies, De Jong et al. (2019) applied TPB to the study of public participation behavior and found that subjective norm had no significant influence on intention. In addition, another study on green consumption also found that subjective norm had no significant influence on intention (Mamun et al., 2018). The results of our study indicate that when people participate in UGS governance, the opinions of their relatives and friends have no significant influence on their intention to participate. This differs slightly from Mathers et al. (2015) conclusions that partnerships may influence citizens' participatory behavior. The reason may be that the social collective participatory governance model of UGS has not been widely promoted in China, and most people's participatory behavior in UGS governance remains highly subjective. For example, the community garden model, which has been well developed in some countries, often serves as a social space for people to establish relationships (Glover et al., 2005). There are also some successful community garden cases in China, which have achieved a good governance effect and enhanced the cohesion of surrounding communities. On the whole, however, this kind of UGS governance mode is relatively short of development space in China. There are three reasons for this. First, there is a lack of urban land to develop community gardens. Especially in highly urbanized big cities, the distribution of urban open spaces is relatively fragmented. Second, the current high-intensity work rhythm leaves young people with limited leisure time, and the majority of people who can participate may be retired. This makes the community gardens relatively lack the participation of young people. Third, most of the current community green space managers are enterprises. However, community gardens have a strong social welfare bias, and it is difficult for them to develop if they are only promoted by enterprises. This requires an opportunity for stakeholders to communicate their real needs to each other, which is what is currently lacking.

The fitting coefficient of the model (Table 4) shows that perceived behavioral control has a strong influence on both intention and behavior, which is a key influence variable in our research model. Moreover, the indirect effect of perceived behavioral control on behavior is small, while the direct effect on behavior is greater, which indicates that the effect of perceived behavioral control on behavior cannot be demonstrated through intention; that is, the path of PBC-B cannot be ignored in the model. In other words, before participating in UGS governance, people will first assess whether they have the objective conditions for participation, such as relevant skills, time, and equipment. The assessment results will have a major impact on the intention to participate and limit actual behavior. It can be observed from the data in Table 2 that the average values of attitude are close to 4,

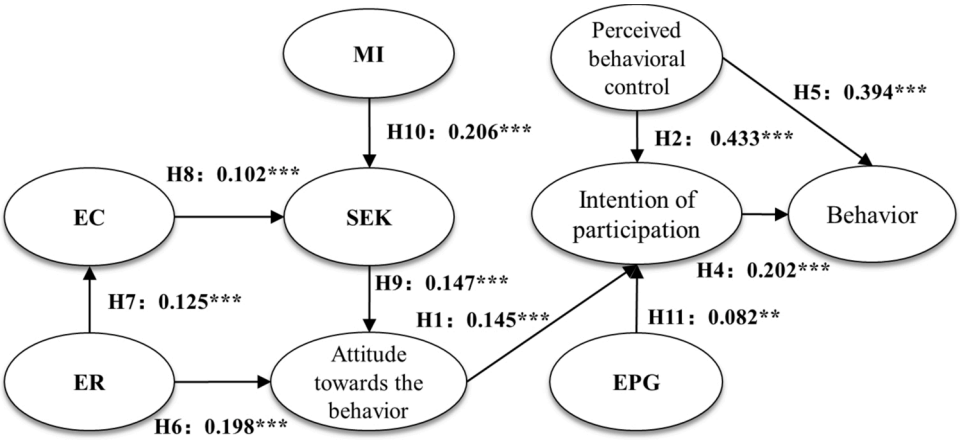


Fig. 3. Revised research model.

the average values of intention are between 3 and 4, while the average values of behavior are below 3 (below the general level). This shows that although people are in favor of participating in UGS governance, there are crucial factors that have an impact on the transformation of attitudes into intentions and intentions into actual behaviors, in which people's own conditions account for a large part. This conclusion is consistent with that of Gage and Thapa (2012). The objective conditions of the citizens themselves cannot be changed, but the government can create conditions for more citizens to participate by optimizing participation procedures. For example, diversified participation paths are established so that citizens can choose appropriate participation methods according to their own objective conditions. At the same time, it can also reduce the complexity of the participation procedure and reduce the meaningless time consumption of citizens in the participation process.

Expected personal gratification has a significant impact on intention to participate. As shown in Table 2, the average value of expected personal gratification is approximately 4, indicating that people generally believe that participating in UGS can result in good feelings. However, the model coefficient of expected personal gratification is small, a finding that contrasts with De Jong et al. (2019) findings. In their model, expected personal gratification was the factor that had the greatest impact on intention. However, when they grouped the samples, they found that the coefficient of perceived behavioral control of the digital platform group was significantly higher than that of expected personal gratification. It is not difficult to find that the external conditions of participation (such as mode, limitation, etc.) have an impact on the coefficient between different latent variables. This may be why the actual participation behavior of people in different countries varies, which is due to the environmental construction of public participation under different policy backgrounds. In other words, while maintaining people's current attitudes and expected personal gratification, we can enhance the perceived behavioral control by optimizing the environment of participation, and finally achieve the purpose of improving people's actual participation behavior. In this study, the impact coefficient of expected personal gratification on intention is lower than that of attitude, and attitude is mainly affected by environmental responsibility. This indicates that the current citizens regard participation in the governance of UGS as a kind of behavior that is biased toward public welfare. Having a high intention to participate is more affected by individual ability and attitude, and has a weak relationship with expected personal satisfaction. From this perspective, the majority of current citizens are rational and non-hedonic. Therefore, in addition to citizens' personal ability, attitude is also a key factor.

5.2. Influencing factors of participatory attitude

Our study found that environmental responsibility is the most important factor in the influencing factors of participatory attitude, which can directly and indirectly positively affect citizens' participatory attitudes. According to some previous studies, although there are differences between samples from different countries, the motivation for active participation basically includes "helping the environment" and "learning natural knowledge" (Mathers et al., 2015; Sloane and Pröbstl-Haider, 2019). These two factors can be summarized in this study as environmental responsibility and subjective environmental knowledge. Our study results show that environmental responsibility can not only directly affect citizens' participatory attitude, but also indirectly affect participatory attitudes by increasing environmental concern and promoting citizens' learning of environmental knowledge. Therefore, the question of how to improve the participatory attitude of the public can be transformed into the question of how to increase the environmental responsibility and environmental knowledge of the public. Studies have found that an important reason why people are willing to participate in environmental public welfare activities and provide volunteer services is that they have a sense of local attachment (Williams et al., 1992; Fairweather et al., 2005). Place attachment contains two basic dimensions: place dependence and place identity (Williams et al., 1992). The former tends to be functionally dependent, which mainly refers to facility conditions that can meet people's needs, such as landscape planning, facility level, and pollution treatment. The latter tends to be spiritually dependent, which mainly refers to people's recognition that place is the key element of self-display and social interaction, such as local culture and local activities. This requires local governments to strengthen the construction of these infrastructures and the local cultural atmosphere to enhance citizens' environmental responsibility. On the other hand, the results also show that the increase in environmental knowledge benefits from media publicity. In this regard, compared with traditional media, such as newspapers, outdoor, radio and television, online media, such as popular social software and video websites, seem to have better publicity effect among young people. UGS managers can formulate corresponding publicity methods according to the media preferences of different groups to achieve the desired effect.

5.3. Limitations and future research

Although our research has made a clear contribution, some limitations must be considered. First, similar to other sociological surveys of public participation, it is inevitable that our sample consisted of self-selected citizens. Participation in the questionnaire is also a kind of public participation behavior. It is reasonable to assume that those willing to participate in our study are more likely to participate in UGS governance than those who refused to participate. Participants' demographic data may also be considered not to be fully consistent with the prevailing demographic data. As Beijing is the capital of China, the proportion of those with higher education will be higher in the population composition than in most Chinese cities, which meant that more than half of our sample had a bachelor's degree or above. However, in addition to higher education, our sample did contain other rich demographic data, which was sufficient to support our conclusion.

Second, in our study, we did not classify citizens' participatory behavior in detail, but adopted a broader definition. Nevertheless, the average value of behavior is still below the general level (Table 2); that is, behavior may not occur. Considering that the level of public participation in UGS governance in China is still in its infancy, it may not be able to find enough corresponding grouping samples to develop detailed participatory behavior classification. In fact, the results of this sample collection confirm our initial concerns. Therefore, the types of participatory behavior in this study are mixed, which is more conducive to collecting a wealth of samples for experiments. For future research, UGS governance behaviors can be further subdivided into government-related and government-unrelated behaviors, and this will be more valuable for further research on UGS governance.

Third, in the model constructed, the average values of the two variables (I and PBC) that directly and positively affect participating behaviors are both greater than the general level, but the average value of behavior is less than the general level (Table 2). This indicates that there are still unknown factors that negatively affect participatory behavior. In future studies, more behavioral science theories can be combined to explore how to optimize the model.

6. Conclusions

The purpose of this study was to explore what individual factors may influence citizens' participation in UGS governance and how these individual factors interact with each other; these aspects have not been explored in detail in the previous literature. In this study, SEM was used to test the hypothesis of the extended TPB model. The results demonstrate that apart from SN, which has no significant effect on intention, the other factors are consistent with the initial hypothesis. The conclusion of this study is helpful in enriching the relevant theoretical system of citizens' participation in UGS governance and provides a scientific reference for urban landscape management departments to formulate

effective and targeted public participation policies.

Author statement

Yuan Huang: Conceptualization, Methodology, Writing- Original draft preparation. **Francisco Aguilar**: Writing- Reviewing and Editing. **Jie Yang**: Software, Investigation. **Yueting Qin**: Investigation, Visualization. **Yali Wen**: Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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